

PRINT SHOP OPERATIONS PLATFORM

JobScheduler

Service catalog & instant pricing · MoMo deposits ·
Appointment slots · Production scheduling · Team chat ·
Analytics

System documentation · Tech stack · API reference · Price list (GHS)
github.com/bvggies/jobscheduling · Generated 23 June 2026

JobScheduler

Print shop operations platform — service catalog & instant pricing, MoMo deposits, appointment slots, production scheduling, team chat, and analytics.

Built for a real print business workflow: customers order online with fixed GHS prices, pay an **80% MoMo deposit**, book **system-generated time slots**, and track production until completion.

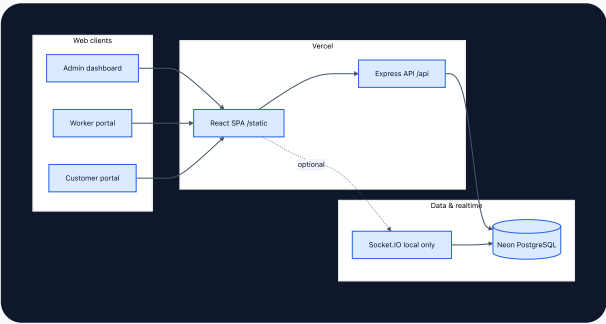
 [Download System Guide \(PDF\)](#) — full visual documentation with diagrams, tech stack, API reference, and price list.

To regenerate the PDF after editing this README: `npm run docs:pdf`

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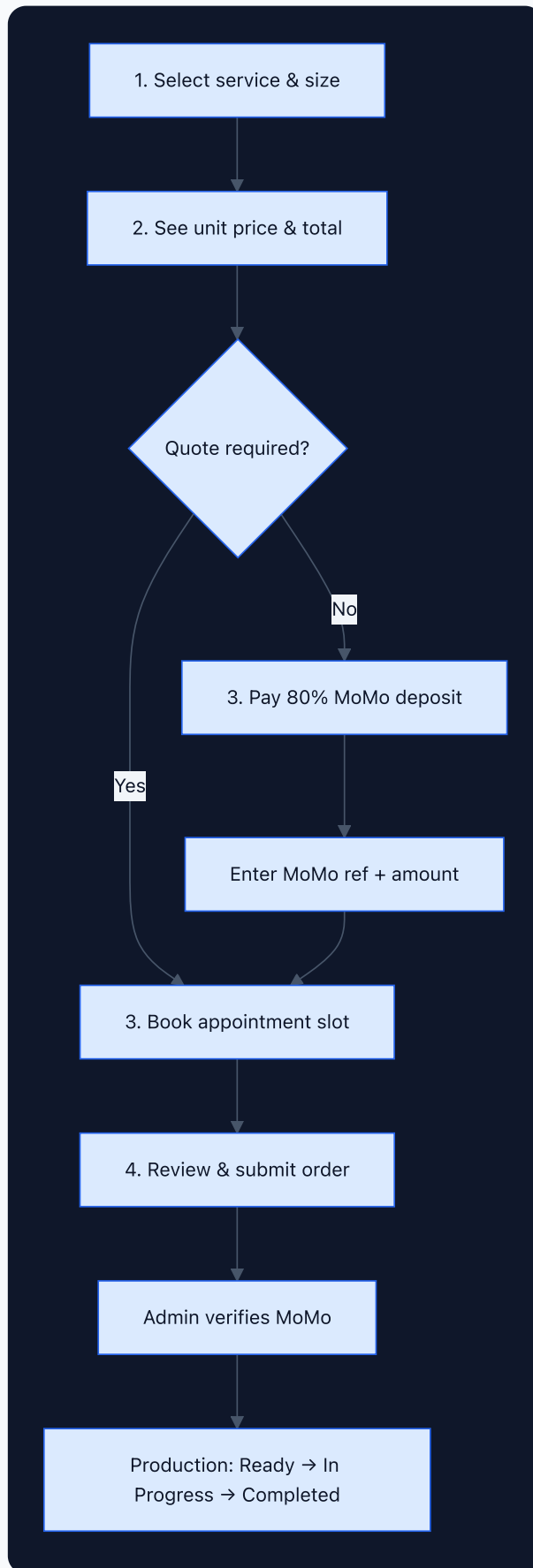
System at a glance



Layer	Technology	Purpose
Frontend	React 19, React Router 7, CRA	Three role-based portals
Backend	Node.js, Express 5	REST API + JWT auth
Database	PostgreSQL (Neon)	Jobs, services, users, chat, alerts
Hosting	Vercel	Static SPA + serverless API
Realtime	Socket.IO	Live chat (long-lived Node host only)

How the business flow works

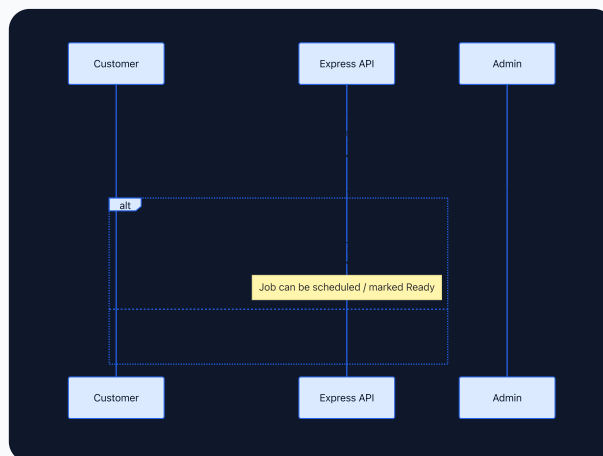
Every customer order follows the same rules:



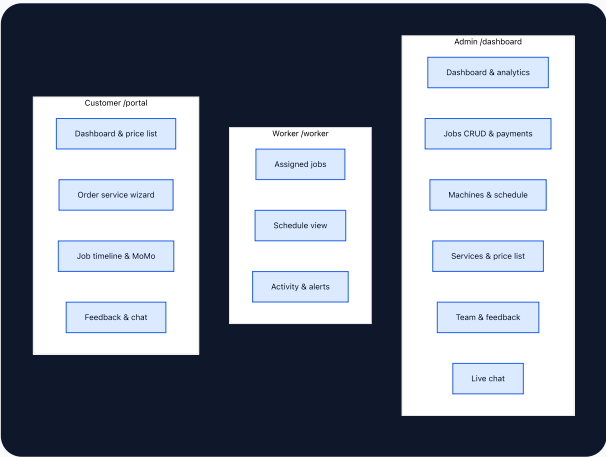
Customer rules (enforced in code)

Rule	Implementation
Instant pricing	Prices loaded from <code>shop_services</code> DB table; total = unit × quantity (or tiered yards)
80% deposit	<code>deposit_required = total × 0.8</code> ; validated on client and server
MoMo before slot	Wizard order: Service → Deposit → Slot → Submit
Slots only	No manual date picker; <code>/api/schedule/available-slots</code>
5 PM cutoff	Last bookable slot ends at 17:00 (same-day rule)
Break excluded	No slots 12:30 – 13:00 daily
Work after deposit	Jobs cannot go Ready or enter auto-scheduler until deposit is Received
Quote services	Digital Flag & Signage: admin sets price → customer pays deposit

MoMo payment flow



User roles & portals



Role	Login	Main capabilities
Admin	/login → /dashboard	Full shop control: jobs, machines, schedule, services/prices, MoMo verification, analytics, team
Worker	/login → /worker	Assigned jobs, status updates, schedule, alerts, chat
Customer	/register or /login → /portal	Order services, pay MoMo deposit, book slots, track jobs, message shop

Admin routes

Path	Feature
/dashboard	KPIs, job overview
/jobs	Job list, search, status
/jobs/new , /jobs/edit/:id	Manual job entry (back office)
/machines	Equipment & substrate compatibility
/schedule	Timeline + drag-and-drop + auto-schedule
/services	Upload/edit service catalog & GHS prices
/analytics	Revenue, utilization, on-time rate
/alerts	Risk, late, rush, underutilization
/team	Customers & workers
/feedback	Customer message threads
/chat	Live team messaging
/activity	Shop-floor activity feed

Customer routes

Path	Feature
/portal	Dashboard, how-it-works, full price list
/portal/jobs/new	Order wizard (service → deposit → slot)
/portal/jobs/:id	Job detail, MoMo status, timeline
/portal/feedback	Feedback to shop
/portal/chat	Live chat

Tech stack

Frontend

Package	Version	Use
React	19.x	UI framework
React Router	7.x	Role-based routing
Axios	1.x	HTTP client + JWT interceptors
date-fns	4.x	Dates & formatting
Framer Motion	12.x	Page transitions
AOS	2.x	Scroll animations
@dnd-kit	6.x	Schedule drag-and-drop
Socket.IO client	4.x	Live chat
React Icons	5.x	Icon set
Create React App	5.x	Build toolchain

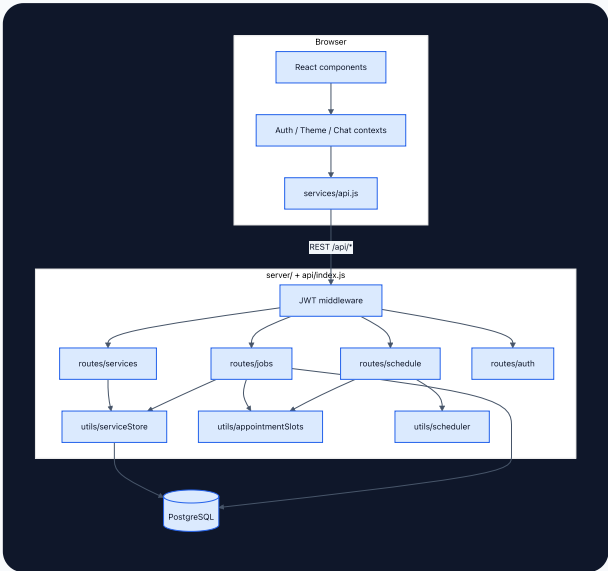
Backend

Package	Version	Use
Express	5.x	HTTP API
pg	8.x	PostgreSQL pool
bcryptjs	3.x	Password hashing
jsonwebtoken	9.x	JWT sessions (7-day)
Socket.IO	4.x	Chat WebSocket
dotenv	17.x	Environment config

Infrastructure

Service	Role
Vercel	Hosts React build + <code>/api</code> serverless functions
Neon	Serverless PostgreSQL with SSL
GitHub	Source control & Vercel deploy trigger

Architecture



Smart scheduling (production)

Auto-scheduler (`server/utils/scheduler.js`):

1. Pick jobs with **deposit received**, no machine assigned
2. Sort by **priority** (Rush first) then **due date**
3. **Group by substrate + finishing** to reduce changeovers
4. Assign first **compatible machine**; slot after last job on that machine
5. Estimate duration from product type + quantity

Manual overrides via `/schedule` drag-and-drop or `PUT /api/schedule/:jobId` .

Authentication

- JWT in `localStorage` (`jobscheduler_token`)

- Roles: `admin` , `worker` , `customer`
- Production fallback: JWT secret derived from `DATABASE_URL` if `JWT_SECRET` unset (set `JWT_SECRET` in prod for best practice)
- Bootstrap admin via `INITIAL_ADMIN_EMAIL` / `INITIAL_ADMIN_PASSWORD` when no admin exists

Service catalog & pricing (GHS)

Prices live in the `shop_services` database table (seeded on first catalog load). Admins edit them at `/services` .

Fixed-size services

Service	Options & prices (GHS)
DTF Printing	A4 ₵5 · A3 ₵10 · A2 ₵20
Banner Printing	3×4 ₵180 · 4×4 ₵200 · 5×4 ₵230 · 6×4 ₵250 · 8×4 ₵300 · 8×8 ₵350
Sticker Printing	A4 ₵5 · A3 ₵10 · A2 ₵15 · 4out ₵20
Wooden Frame	12×16 ₵160 · 15×19 ₵200 · 19×23 ₵300 · 20×30 ₵450 · 24×36 ₵600
T-Shirt Printing	Jersey ₵50 · Silk ₵55 · Cotton ₵60 · Lacoste ₵70

Per square foot

Service	Price
Sticker	₵3.0 / sq ft
Banner	₵3.5 / sq ft
One-way vision	₵6.0 / sq ft
Reflective sticker	₵8.0 / sq ft

Tiered

Service	Tiers
Material — Cloth Printing	1–49 yards ₵45/yd · 50–99 ₵43/yd · 100+ ₵40/yd

Quote on request

Service	Notes
Digital Flag Printing	Available now — admin sets quote
Signage / 3D Signboard	Price depends on size and letters

Deposit rule: all priced orders require **80% of total** via MoMo before production starts.

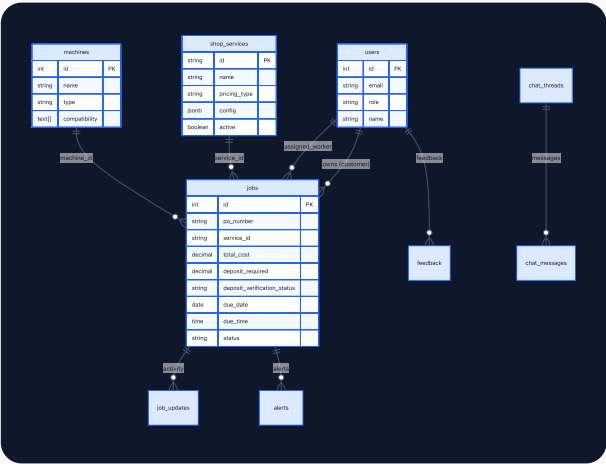
Shop hours & contact

Mon – Thu	7:30 AM – 6:00 PM
Friday	7:30 AM – 7:00 PM
Saturday	10:00 AM – 6:00 PM
Sunday	Closed
Break	Daily 12:30 PM – 1:00 PM
Same-day cutoff	Send work by 5:00 PM latest

Contact	
Call	0542917636 / 0207924080
WhatsApp	0542917636
MoMo	05429176360 — Tetteh Henry

Database overview

Schema is applied via `server/apply-schema.js` (no separate SQL migration files).



Key jobs payment fields

Column	Description
total_cost	Order total (GHS)
deposit_required	80% of total
deposit_received	Confirmed amount
deposit_status	Pending / Received
deposit_momo_phone	Customer MoMo number
deposit_momo_reference	Transaction ID
deposit_submitted_amount	Amount customer paid
deposit_verification_status	none / pending / confirmed / rejected

API reference

Base URL: {host}/api — all protected routes need Authorization: Bearer <token> .

Auth

Method	Path	Description
POST	/auth/register	Customer registration
POST	/auth/login	Login (all roles)
GET	/auth/me	Current user profile

Services & catalog

Method	Path	Auth	Description
GET	/services/catalog	Public	Active services, price list, hours, contact
POST	/services/calculate	Public	Price calculation
GET	/services/manage	Admin	All services
POST	/services/manage	Admin	Create service
PUT	/services/manage/:id	Admin	Update service
DELETE	/services/manage/:id	Admin	Deactivate service

Jobs

Method	Path	Auth	Description
GET	/jobs	All	List jobs (scoped by role)
POST	/jobs	Customer/Admin	Create job (customer = catalog flow)
GET	/jobs/:id	All	Job detail
PUT	/jobs/:id	Admin	Update job
PATCH	/jobs/:id/status	Admin/Worker	Status change
PATCH	/jobs/:id/submit-deposit	Customer	Submit MoMo payment
PATCH	/jobs/:id/verify-deposit	Admin	Confirm/reject MoMo
PATCH	/jobs/:id/quote	Admin	Set quote pricing
PATCH	/jobs/:id/payment	Admin	Record deposit/final payment

Schedule

Method	Path	Auth	Description
GET	/schedule/available-slots	All	Bookable appointment slots
GET	/schedule	Admin/Worker	Production schedule
POST	/schedule/auto-schedule	Admin	Run auto-scheduler
PUT	/schedule/:jobId	Admin	Manual schedule update

Other

Prefix	Features
/machines	Equipment CRUD
/analytics	Admin reports
/alerts	Notifications
/users	Team management
/feedback	Customer feedback
/activity	Job activity feed
/chat	Messaging threads
GET /health	API health check

Project structure

```

jobscheduling-1/
├── api/
│   └── index.js          # Vercel serverless entry → server/index.js
├── public/               # Static assets, logo, manifest
├── src/
│   ├── components/
│   │   ├── ServiceOrderForm.js # Customer order wizard
│   │   ├── DepositPaymentForm.js
│   │   ├── ShopInfoPanel.js
│   │   ├── JobPayment.js       # Admin MoMo verify + quotes
│   │   └── Layout/            # Navbar, Sidebar
│   ├── context/             # Auth, Theme, ChatSocket
│   ├── hooks/
│   │   └── useServiceCatalog.js
│   ├── layouts/             # Admin, Worker, Customer shells
│   ├── pages/
│   │   ├── customer/         # Customer portal pages
│   │   ├── worker/           # Worker portal pages
│   │   └── ServiceManagement.js # Admin price list editor
│   ├── services/api.js      # Axios + API modules
│   └── utils/
│       ├── shopConfig.js     # Hours, contact, 80% deposit
│       └── servicePricing.js  # Client-side price math
├── server/
│   ├── config/database.js    # Pool + schema init
│   ├── routes/               # Express route modules
│   ├── utils/
│   │   ├── serviceStore.js   # DB-backed catalog
│   │   ├── appointmentSlots.js # Slot generation (5 PM cutoff)
│   │   ├── scheduler.js      # Auto-scheduler
│   │   └── catalogData.js     # Default seed prices
│   ├── apply-schema.js       # Run migrations
│   ├── index.js              # Express app
├── vercel.json               # Vercel builds & routes
└── package.json

```

Local development

Prerequisites

- Node.js 18+ (20+ recommended)
- PostgreSQL (Neon connection string or local)
- npm

Install & run

```
# Clone
git clone https://github.com/bvggies/jobscheduling.git
cd jobscheduling

# Frontend
npm install

# Backend
cd server && npm install && cd ..

# Environment - root .env
echo 'REACT_APP_API_URL=http://localhost:5000/api' > .env

# Environment - server/.env
cat > server/.env << 'EOF'
DATABASE_URL=postgresql://user:pass@host/db?sslmode=require
PORT=5000
NODE_ENV=development
JWT_SECRET=dev-secret-change-me
INITIAL_ADMIN_EMAIL=admin@yourshop.local
INITIAL_ADMIN_PASSWORD=YourSecurePass123
EOF

# Apply schema
cd server && node apply-schema.js && cd ..

# Terminal 1 - API
npm run server

# Terminal 2 - React
npm start
```

Open <http://localhost:3000>

Script	Command	Description
Frontend dev	<code>npm start</code>	CRA dev server :3000
Backend	<code>npm run server</code>	Express + Socket.IO :5000
Backend watch	<code>npm run server:dev</code>	Nodemon
Production build	<code>npm run build</code>	Output to <code>build/</code>
Schema	<code>cd server && node apply-schema.js</code>	Apply DB schema

Database migrations

There are no versioned SQL migration files. Schema is managed in `server/config/database.js` and applied with:


```
cd server
DATABASE_URL="your-neon-connection-string" node apply-schema.js
```

This creates tables, indexes, default machines, and bootstraps an admin user if none exists.

Deployment (Vercel + Neon)



1. Create a [Neon](#) project and copy the connection string
2. Import repo in [Vercel](#)
3. Set environment variables (see below)
4. Deploy — run `apply-schema.js` against Neon once
5. Set `REACT_APP_API_URL=https://your-app.vercel.app/api` and redeploy

See also: [DEPLOYMENT.md](#) · [VERCEL_DEPLOY.md](#) · [SETUP_ENV.md](#)

Socket.IO on Vercel

Vercel serverless **cannot host WebSockets**. Live chat uses HTTP on Vercel by default. For realtime chat, set `REACT_APP_WS_ORIGIN` to a long-lived Node host running `server/index.js`.

Environment variables

Vercel / production

Variable	Required	Description
DATABASE_URL	Yes	Neon PostgreSQL connection string
NODE_ENV	Yes	production
JWT_SECRET	Recommended	Signing key for JWT
REACT_APP_API_URL	Yes	e.g. https://your-app.vercel.app/api
INITIAL_ADMIN_EMAIL	Optional	First admin bootstrap email
INITIAL_ADMIN_PASSWORD	Optional	First admin bootstrap password
REACT_APP_WS_ORIGIN	Optional	Socket.IO host for live chat
REACT_APP_CHAT_SOCKET	Optional	true / false to force chat socket

Local (server/.env)

Variable	Description
DATABASE_URL	Postgres connection
PORT	API port (default 5000)
JWT_SECRET	JWT signing secret

Local (root .env)

Variable	Description
REACT_APP_API_URL	http://localhost:5000/api

Related docs

Document	Contents
JobScheduler-System-Guide.pdf	Full visual PDF — diagrams, tech stack, flows, API, pricing
QUICKSTART.md	Fast setup guide
DEPLOYMENT.md	Full deploy walkthrough
VERCEL_DEPLOY.md	Vercel-specific notes
SETUP_ENV.md	Environment configuration
TROUBLESHOOTING.md	Common issues
START_SERVER.md	Running the API locally

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Support

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